**Modélisation, estimation, simulation des risques climatiques**
**Session 3**

# Scenario building of Integrated Assessment Models

Gauthier Vermandel
Ecole polytechnique

# Schedule of the day

## Objectives

- Understanding the **basics** of **Integrated Assessment Models**.
- Build and **simulating macro-climate scenarios** using IPCC and NGFS methodology.

## Materials:

- notebook_session3.ipynb *notebook introduction of model's options.*
- notebook_session3.ipynb *notebook exercises about scenarii simulations.*
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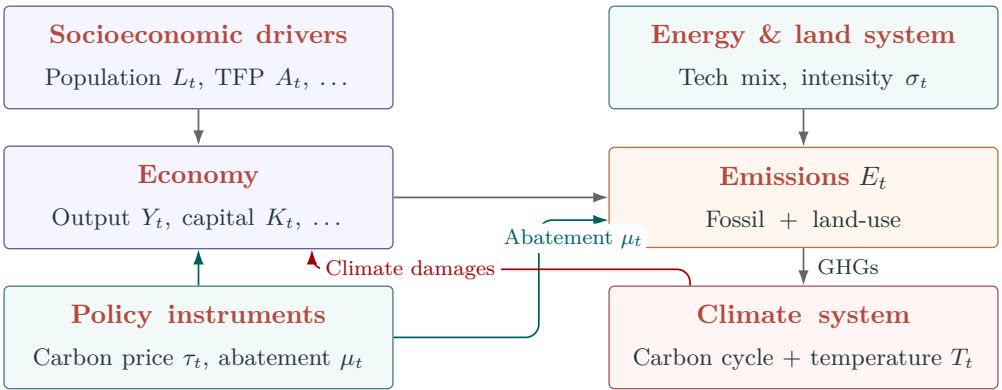
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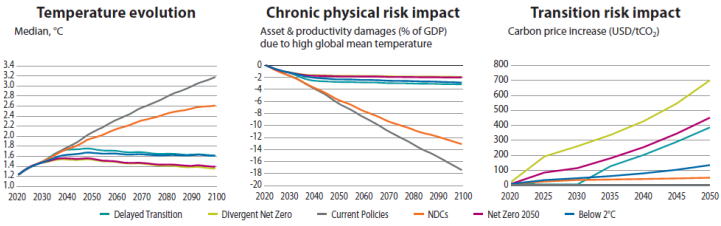
# Introduction

# Introduction: why IAMs matter for policy and finance



# Introduction: IAMs in practice: NGFS Scenarios

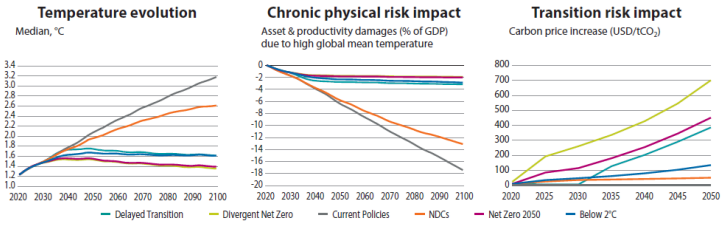
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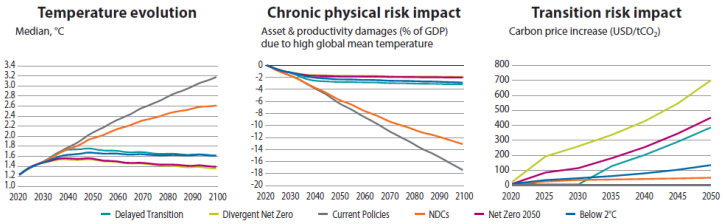
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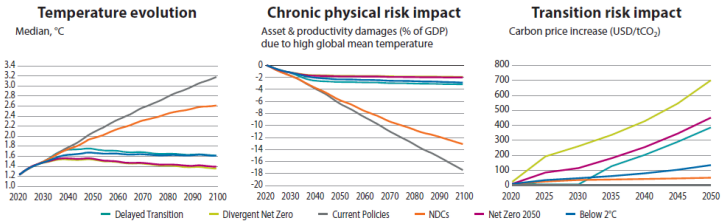
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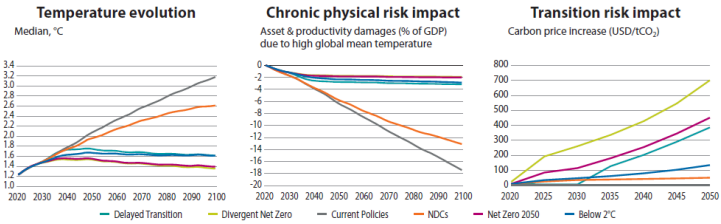
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- They differ because of two main ingredients:
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     - Energy and land-use technology adoption (renewables, CCS, etc.).
     - Speed of structural change and decarbonization potential.
  2. **Policy implementation (forcing levels)**
     - Presence (or absence) of a carbon price.
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# Outline

1 Introduction

2 Shared Socioeconomic Pathways (SSPs)

3 The DICE Model

- The socioeconomic block
- Economic decisions

# Shared Socioeconomic Pathways (SSPs)

# SSPs: Key definitions

- **SSPs (Shared Socioeconomic Pathways):** Narratives + quantitative pathways describing alternative futures of population, GDP, technology, land use, energy.
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- **SSP2 – Middle of the road**: historical trends continue.
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# SSPs: The 5 SSP narratives

- **SSP1 – Sustainability**: inclusive development, green growth.
- **SSP2 – Middle of the road**: historical trends continue.
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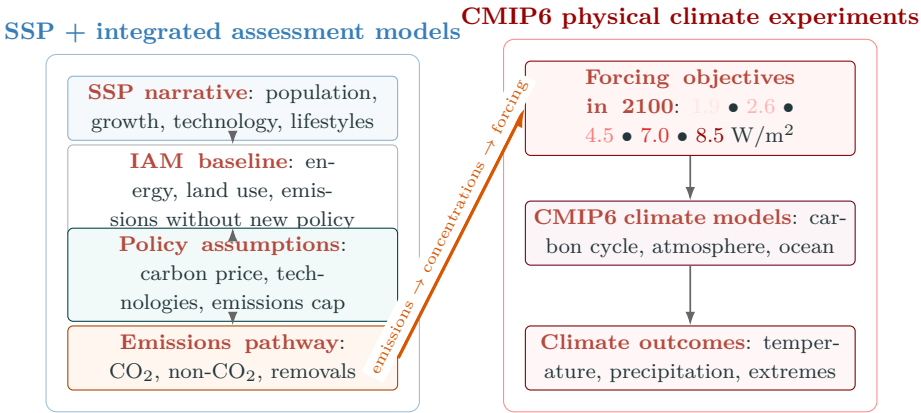
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# SSPs: The SSP × forcing matrix

| | Forcing $W/m^2$ | | | | |
|---|---|---|---|---|---|
| | 1.9 | 2.6 | 4.5 | 7.0 | 8.5 |
| SSP1 | SSP1-1.9 | SSP1-2.6 | – | – | – |
| SSP2 | – | SSP2-2.6 | SSP2-4.5 | SSP2-7.0 | – |
| SSP3 | – | – | SSP3-4.5 | SSP3-7.0 | SSP3-8.5 |
| SSP4 | – | SSP4-2.6 | SSP4-4.5 | SSP4-7.0 | – |
| SSP5 | – | – | SSP5-4.5 | – | SSP5-8.5 |

Each cell = a scenario used in IPCC/CMIP6 (e.g., **SSP1-2.6**, **SSP5-8.5**).

# From an SSP storyline to a climate outcome




**Key point:** the forcing label is an *outcome to be reached*, not a climate trajectory mechanically attached to an SSP. IAMs must find a policy and technology pathway consistent with both.

# Why are some SSP–forcing combinations missing?

## A combination must pass three tests

1. **Narrative consistency**: does the policy path still resemble the SSP?
2. **Model feasibility**: can IAMs reach the target with available technologies and resources?
3. **Scenario selection**: was it retained for CMIP6 experiments?

## Example: SSP5 with low forcing

SSP5 starts from rapid, fossil-intensive growth. Reaching 1.9 or 2.6 $W/m^2$ would require:

- a very high and rapid carbon constraint;
- accelerated clean-technology substitution;
- often large-scale carbon removal and net-negative emissions.

**A blank cell does not automatically mean "physically impossible."** It may mean *not selected*, *not robustly produced by IAMs*, or *attainable only under assumptions that strain the SSP narrative.*

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# SSPs: Two contrasted futures

**SSP1-2.6 (Sustainability + strong mitigation)**

- Low population, inclusive growth.
- Rapid clean tech diffusion.
- $CO_2$ emissions peak early, net zero 2070.
- Warming: +1.5–2°C.

**SSP5-8.5 (Fossil-fueled growth, no policy)**

- High GDP, energy-intensive growth.
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# The DICE Model

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- DICE (Dynamic Integrated Climate-Economy model) was developed by Nordhaus (1992) and went through several (minor) updates up to the lastest 2023 snapshot;
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# DICE: setup basics

- Time is discretized in steps $i = 1, \ldots, I$.
  - calendar time: $t(i) = t_0 + \Delta \cdot i$
  - $t_0$ initial period, $\Delta$ time step (in years)
  - horizon: $t \in \{t_0,\, t_0 + \Delta,\, t_0 + 2 \cdot \Delta, \ldots, T\}$
- Variables can be indexed either by *time* or by *step*:

$$Y_t \quad \equiv \quad Y(i) \qquad \textit{with} \quad t = t(i).$$

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# DICE: Carbon cycle

- Atmospheric loading of $CO_2$ is a system of differential equations:

$$M_t = \Phi_M M_{t-\Delta} + \Xi_M E_t, \tag{1}$$

where $M_t = [M_t^{AT}, M_t^{UP}, M_t^{LO}]'$ comprises 3 layers of carbon: atmospheric, upper and lower ocean respectively.

- Matrices are determined by:

$$\Phi_M = \begin{bmatrix} 1-\Delta b_{12} & \Delta b_{12}\frac{M_{AT0}}{M_{UP0}} & 0 \\ \Delta b_{12} & 1-\Delta b_{12}\frac{M_{AT0}}{M_{UP0}} - \Delta b_{23} & \Delta b_{23}\frac{M_{UP0}}{M_{LO0}} \\ 0 & \Delta b_{23} & 1-\Delta b_{23}\frac{M_{UP0}}{M_{LO0}} \end{bmatrix} \text{ and } \Xi_M = \begin{bmatrix} \xi \\ 0 \\ 0 \end{bmatrix}$$

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$$\Phi_T = \begin{bmatrix} (1 - \Delta \cdot c_1 \cdot F_{2\times CO2} - \Delta \cdot c_1 \cdot c_3) & (\Delta \cdot c_1 \cdot c_3) \\ (\Delta \cdot c_4) & (1 - \Delta \cdot c_4) \end{bmatrix} \text{ and } \Xi_T = \begin{bmatrix} \Delta \cdot c_1 \\ 0 \end{bmatrix}$$

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# DICE: Radiative forcing

- Greenhouse effect via radiative forcing ($\mathrm{W/m^2}$):

$$F_t = F_{2\times CO_2} \ \log\left(M_{t-\Delta}^{AT}/M_{AT0}\right)/\log(2) + F_t^{EX} \tag{3}$$

- Here, $F_{2\times CO_2}$ denote the forcing (Wm-2) of equilibrium CO2 doubling ($M_t^{AT} = 2M_{AT0}$).
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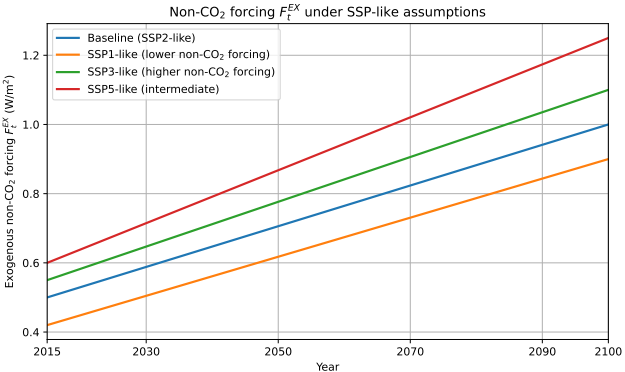
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# DICE: Example of exogenous forcing



# DICE: Input structure

- World population (millions):

$$L_t = L_{t-\Delta} \left(L_\infty / L_{t-\Delta}\right)^{\ell_g}, \tag{4}$$

$\ell_g \in [0, 1]$ convergence rate, $L_\infty \in [0, +\infty)$ long term population.

- Law of motion of capital:

$$K_t = (1 - \delta_K)^\Delta K_{t-\Delta} + I_t, \tag{5}$$

$\delta_K \in [0, 1]$ is the depreciation rate, while $\Delta$ is the time step (here $\Delta = 5$ years)

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- World population (millions):

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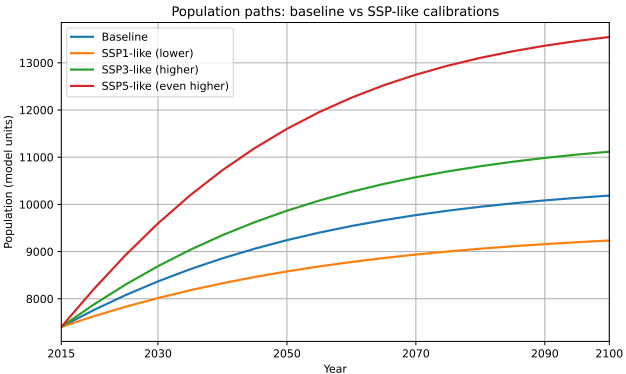
  $\ell_g \in [0, 1]$ convergence rate, $L_\infty \in [0, +\infty)$ long term population.

- Law of motion of capital:

$$K_t = (1 - \delta_K)^{\Delta} K_{t-\Delta} + I_t, \tag{5}$$

  $\delta_K \in [0, 1]$ is the depreciation rate, while $\Delta$ is the time step (here $\Delta = 5$ years)

# DICE: Example of population dynamics



# DICE: Production of goods

- Production is Cobb-Douglas in thousands USD:

$$Y_t = A_t K_{t-\Delta}^{\gamma} \left( \frac{L_t}{1,000} \right)^{1-\gamma} \tag{6}$$

where $\gamma \in [0, 1]$ is capital intensity, physical capital $K_{t-\Delta}$.

- Exogenous total productivity $A_t$ (with vanishing growth)

$$A_t = A_{t-\Delta} / (1 - g_A \exp(-\delta_A (t - t_0))) \tag{7}$$

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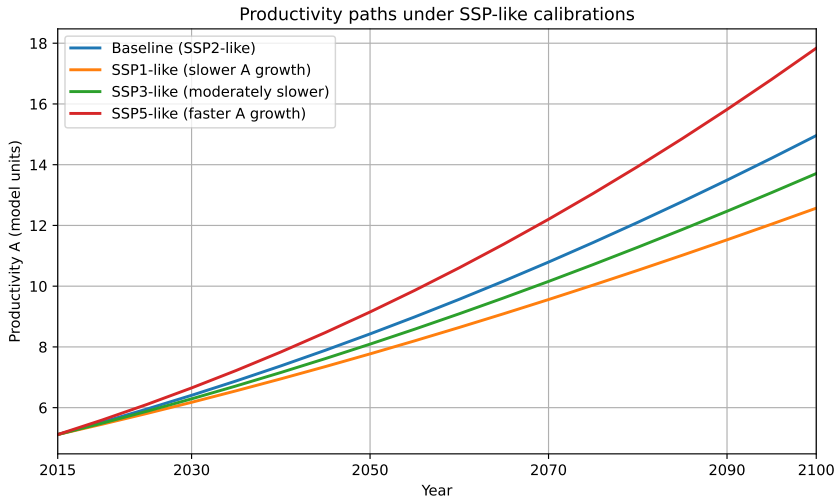
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# DICE: Example of productivity dynamics



# DICE: Emissions

- Emissions (GtCO2):

$$E_t = \Delta \cdot \Big( \sigma_t \left(1 - \mu_t\right) Y_t + E_t^{\text{land}} \Big) \tag{8}$$

where $\sigma_t$ decoupling rate, $\mu_t$ is abatement share, $E_t^{\text{land}}$ is exogenous process of carbon emission from change in land use (i.e. deforestation, ~10% of emissions).

- Exogenous decoupling of emissions to GDP

$$\sigma_t = \sigma_{t-\Delta} \exp\Big( - \; g_\sigma \left(1 - \delta_\sigma\right)^{(t - t_0)} \Delta \Big) \tag{9}$$

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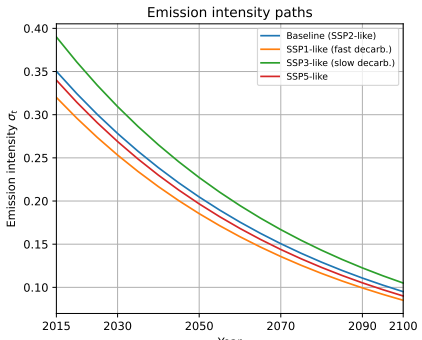
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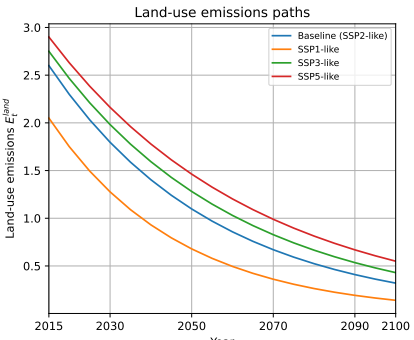
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# DICE: Example of technology dynamics



# DICE: Equilibrium in goods and climate

- Resource constraint:

$$Y_t \left(1 - \theta_{1,t} \mu_t^{\theta_2}\right) \Omega \left(T_{t-\Delta}^{AT}\right) = C_t + I_t \tag{10}$$

LHS is output loss from abatement and damage, RHS is demand side.

- Damages are given by:

$$\Omega \left(T_{t-\Delta}^{AT}\right) = 1/(1 + a \left(T_{t-\Delta}^{AT}\right)^2) \tag{11}$$

This expression is subject to deep uncertainty, and should be taken as a normative measure of the social loss for the society of rising temperatures. Tipping points could be introduced [Weitzman, 2012].

- Abatement good technology decreases over time:

$$\theta_{1,t} = \frac{p_b}{1000\,\theta_2} \left(1 - \delta_{pb}\right)^{(t-t_0)} \sigma_t. \tag{12}$$

where $p_b > 0$ price of carbon to reach net zero, efficiency factor $\delta_{pb} \in [0, 1]$.

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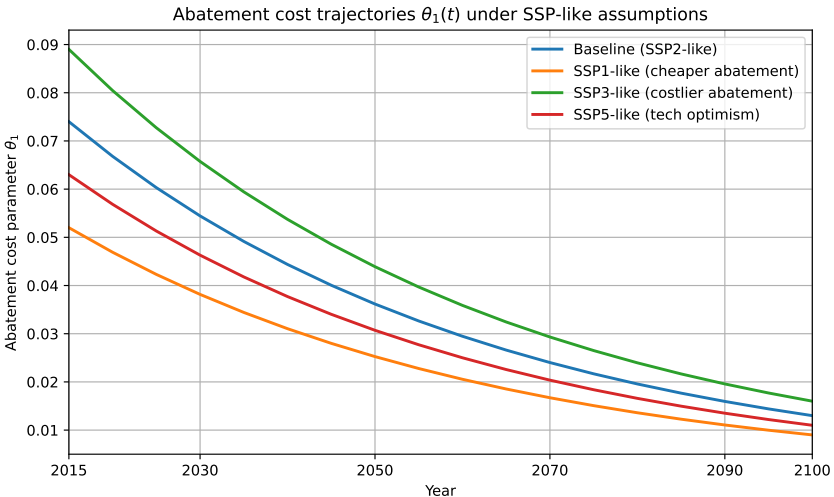
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# DICE: Example of abatement technology



Total factor productivity:

$$A_t = A_{t-\Delta}/(1 - g_A \exp(-\delta_A (t - t_0)))$$

Decoupling rate of carbon emission to GDP:

$$\sigma_t = \sigma_{t-\Delta} \exp\Big( - g_\sigma (1 - \delta_\sigma)^{(t-t_0)} \Delta\Big)$$

Cost of abating carbon emissions:

$$\theta_{1,t} = \frac{p_b}{1000\,\theta_2} (1 - \delta_{pb})^{(t-t_0)} \sigma_t$$

Non-CO2 forcing law of motion:

$$F_t^{EX} = \min\left(F_{t-\Delta}^{EX} + \Delta \cdot d_F, F_{CAP}\right)$$

Land-use law of motion:

$$E_t^{\text{land}} = (1 - \delta_e)^{\Delta} E_{t-\Delta}^{\text{land}}$$

World population dynamic:

$$L_t = L_{t-\Delta} (L_T/L_{t-\Delta})^{\ell_g}$$

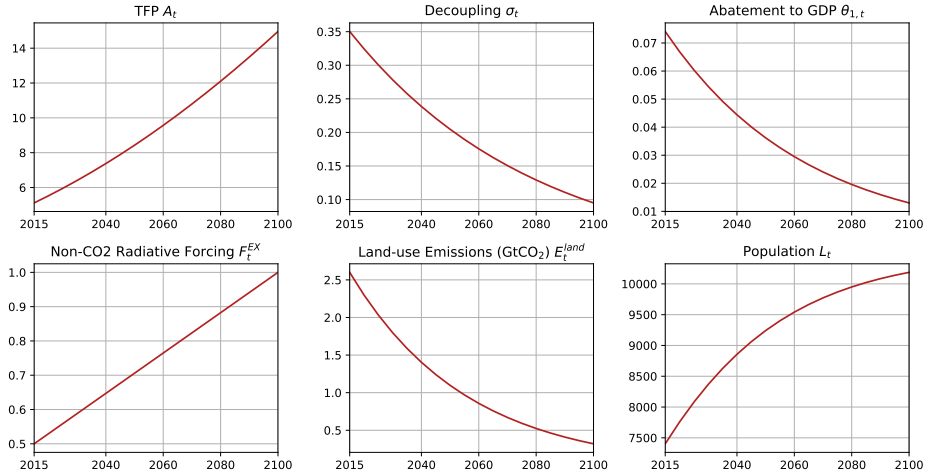



Figure. Summary of exogenous variables

# DICE: Introducing a saving rate

- How should we determine optimally $C$ and $I$?
- We introduce optimal saving rate $s$, consumption and investment becomes:

$$C_t = \Delta \cdot (1 - s_t) \times Y_t \left(1 - \theta_{1,t} \mu_t^{\theta_2}\right) \Omega \left(T_{t-\Delta}^{AT}\right) \tag{13}$$

$$I_t = \Delta \cdot s_t \times Y_t \left(1 - \theta_{1,t} \mu_t^{\theta_2}\right) \Omega \left(T_{t-\Delta}^{AT}\right) \tag{14}$$

  $\rightarrow$implicitely assuming that income spent between $C_t$ and $I_t$.

- Here, $s$ is "decision" variable, which usually is pinned down by optimal control problem in economics.

**Assumption (1)**
*In what follows, we assume that* $\{s_t\}_{t=t_0}^{T} = 0.25$ *as simplifying assumption.*

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# DICE: Introducing the abatement rate

- Environmental policy here, denoted by abatement share $\mu_t$, is the second decision variable.
- Increasing $\mu_t$ has several economic and social implications:
  - $\partial C_t/\partial \mu_t < 0$ : $\mu_t \nearrow$ crowds out **consumption**. Yellow vest effect which can threaten the implementation of the policy.
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# DICE: model summary

- **Exogenous variables** each driven by an exogenous process (growth/decline path).

$$x_t = \{A_t,\ \sigma_t,\ \theta_{1,t},\ F_t^{EX},\ E_t^{\text{land}},\ L_t\}$$

- **Endogenous state/flow variables**

$$y_t = \{M_t^{AT},\ M_t^{UP},\ M_t^{LO},\ T_t^{AT},\ T_t^{LO},\ Y_t,\ K_t,\ C_t\}$$

- **Decision variables (or controls)**

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# DICE: A sketch of the numerical solution

Under Assumptions (1) and (2), the numerical solution is straightforward through an open loop:

$$x_t = g(x_{t-1}), \qquad g: \mathbb{R}^{N_x} \to \mathbb{R}^{N_x},$$
$$y_t = f(y_{t-1}, x_t, z_t), \qquad f: \mathbb{R}^{N_y} \times \mathbb{R}^{N_x} \times \mathbb{R}^2 \to \mathbb{R}^{N_y}.$$

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Thank you!

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